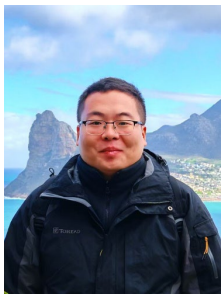
ZE ZHAO (赵泽)



Postdoctoral Fellow | Microbial Ecology, Environmental Microbiology

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Westlake University, Xihu District, Hangzhou 310030, Zhejiang, China

EDUCATION

Ph.D. in Environmental Science and Engineering
Zhejiang University & Westlake University Joint Program | 2020-2025
Thesis: Exploring Assembly Mechanisms of Microbial Communities and Ecological Functions of a Novel Ammonia-Oxidizing Archaea Clade in Intertidal Wetlands

M.S. in Environmental Science and Engineering
School of Marine Science and Technology, Tianjin University | 2017-2020
Thesis: Pollution characteristics and influencing factors of extracellular/intracellular antibiotic resistance genes in Haihe River estuary, Tianjin, China

B.S. in Environmental Science
School of Environmental Science and Engineering, Tianjin University | 2013-2017

RESEARCH EXPERIENCE

- **Postdoctoral Research Associate**
  School of Engineering, Westlake University, China | June 2025-now
  Project: Enrichment and isolation of *Nitrosomaritimum*-like ammonia-oxidizing archaea (AOA) from coastal wetlands and their adaptation mechanisms to oxygen-limited environments.

RESEARCH INTERESTS

Microbial ecology and microbial-driven nitrogen cycles in coastal ecosystems and freshwater lakes, explored through high-throughput sequencing, metagenomics, metatranscriptomics, phylogenomics, microbial enrichment, laboratory experiments, etc.

KEY ACHIEVEMENTS

1. **Discovery of a novel genus of ammonia-oxidizing archaea (AOA) in intertidal wetlands**
   - A novel AOA genus, *Candidatus* Nitrosomaritimum, firstly identified in intertidal aquifers, which possesses genetic potential for urea utilization and anoxic nitrous oxide reduction. This genus is widely distributed in estuarine-coastal zones and marine sediments.
   - Demonstrated that *Nitrososphaeria* genomes selectively encode either the Pst or Phn transporter for phosphate uptake.
   - Revealed that intertidal archaeal *Hydrothermarchaeota* facilitate cooperative detoxification of extracellular hydroxylamine produced by co-occurring AOA.

2. **Microbiota rhythmicity and assembly patterns in the intertidal groundwater-coastal seawater continuum**
   - Uncovered rhythmic patterns of microbial community structure in response to periodic tidal fluctuations in the intertidal groundwater-coastal seawater continuum.
   - Revealed that microbiota assembly in the intertidal water continuum is primarily governed by homogeneous selection, dispersal limitation and drift.

TECHNICAL SKILLS

- **Bioinformatics:**
  - Amplicon analysis: microbial diversity and composition, community assembly, co-occurrence networks, random forest, etc.
  - Metagenomic assembly/binning, functional annotation, phylogenomics, etc.
  - Metatranscriptomic analysis.
- **Tools:** R, Linux/Shell.
- **Wet Lab:** groundwater/sediment sampling, DNA/RNA extraction, qPCR, geochemical analysis, microbial enrichment/isolation culture etc.

AWARDS & HONORS

2023-2024 Westlake University Suwu Scholarship
2023-2024 Westlake University “Innovation Award”
2022-2023 Zhejiang University “Excellent Graduate Student”
2022-2023 Westlake University Muyuan Scholarship for the 2023
2017 Outstanding Graduates of Tianjin University in 2017

SELECTED PUBLICATIONS

Google Scholar Citation and H-index: 1087 / 11
Web of Science Citation and H-index: 805 / 10

(1) **Zhao, Z.**; Gao, H.; Yang, Y.; Deng, Y.; Ju, F*. Fungi as a Critical Component of Lake Microbiota in Response to Cyanobacterial Harmful Algal Blooms. ***Environmental Science & Technology*** **2025**, 59 (22), 11167–11180.

(2) **Zhao, Z.**; Qin, W.; Li, L.; Zhao, H.; Ju, F*. Discovery of Candidatus Nitrosomaritimum as a New Genus of Ammonia-Oxidizing Archaea Widespread in Anoxic Saltmarsh Intertidal Aquifers. ***Environmental Science & Technology*** **2024**, 58 (36), 16040–16054.

(3) **Zhao, Z.**; Zhang, L.; Zhang, G.; Gao, H.; Chen, X.; Li, L.; Ju, F*. Hydrodynamic and Anthropogenic Disturbances Co-Shape Microbiota Rhythmicity and Community Assembly within Intertidal Groundwater-Surface Water Continuum. ***Water Research*** **2023**, 242, 120236.

(4) **Zhao, Z.**; Zhang, K.; Wu, N.; Li, W.; Xu, W.; Zhang, Y.; Niu, Z*. Estuarine Sediments Are Key Hotspots of Intracellular and Extracellular Antibiotic Resistance Genes: A High-Throughput Analysis in Haihe Estuary in China. ***Environment International*** **2020**, 135, 105385.

(5) Gao, H.; **Zhao, Z.**; Zhang, L.; Ju, F*. Cyanopeptides Restriction and Degradation Co-Mediate Microbiota Assembly during a Freshwater Cyanobacterial Harmful Algal Bloom (CyanoHAB). ***Water Research* 2022**, 220, 118674.

(6) Zhu, X.; Zhu, S.; **Zhao, Z.**; Kang, X.; Ju, F*. Microbiome Dynamics during Anaerobic Digestion of Food Waste and the Genetic Potential for Poly (Lactic Acid) Co-Digestion. ***Chemical Engineering Journal* 2023**, 473, 145194.

(7) Zhang, K.; Xin, R.; **Zhao, Z.**; Li, W.; Wang, Y.; Wang, Q.; Niu, Z.*; Zhang, Y*. Mobile Genetic Elements Are the Major Driver of High Antibiotic Resistance Genes Abundance in the Upper Reaches of Huaihe River Basin. ***Journal of Hazardous Materials* 2021**, 401, 123271.

(8) Wu, N.; Zhang, Y.; **Zhao, Z.**; He, J.; Li, W.; Li, J.; Xu, W.; Ma, Y.; Niu, Z*. Colonization Characteristics of Bacterial Communities on Microplastics Compared with Ambient Environments (Water and Sediment) in Haihe Estuary. ***Science of The Total Environment* 2020**, 708, 134876.

(9) Zhang, K.; Xin, R.; **Zhao, Z.**; Ma, Y.; Zhang, Y.; Niu, Z*. Antibiotic Resistance Genes in Drinking Water of China: Occurrence, Distribution and Influencing Factors. ***Ecotoxicology and Environmental Safety 2020**, 188, 109837*.

(10) Yan, H.; Zhang, Y.; Zhang, Z.; **Zhao, Z.**; Zhang, L.; Ju, F. Soil pH and Nitrate Shape Deterministic Assembly of Microbial Communities in Agricultural Soils via *Nitrososphaeria*. **Applied and Environmental Microbiology 2025**, e02067-25.

(11) Yuan, L.; Yu, P.; Huang, X.; **Zhao, Z.**; Chen, L.; Ju, F. Seasonal Succession, Host Associations and Biochemical Roles of Aquatic Viruses in a Eutrophic Lake Plagued by Cyanobacterial Blooms. ***Environment International* 2024**, 109125.

(12) Zhang, Y.; Mao, K.; Chen, K.; **Zhao, Z.**; Ju, F. Symbiont Community Assembly Shaped by Insecticide Exposure and Feedback on Insecticide Resistance of Spodoptera Frugiperda. ***Communications Biology* 2024**, 7 (1), 1–12.

(13) Wang, H.; Wang, Y.; Zhang, G.; **Zhao, Z.**; Ju, F. Temporal Dynamics and Performance Association of the Tetrasphaera-Enriched Microbiome for Enhanced Biological Phosphorus Removal. ***Engineering* 2023**, 29, 168–178.

CONFERENCE PRESENTATIONS

1. Oral presentation at the "**Joint Conference of the 1st ISME Asia Forum and the 2025 Annual Academic Meeting of the Chinese Association for Microbial Ecology**". Hangzhou. 2025.10. (Title: The Hidden Roles of Cyanopeptides and Fungi in Microbial Community Assembly during Lake Cyanobacterial Blooms).

2. Oral presentation at the "**The Sixth Meeting of Bergey's International Society for Microbial Systematics (BISMIS)**". Qingdao. 2025.10. (Title: *Ca.* Nitrosomaritimum: an overlooked genus of ammonia-oxidizing archaea in intertidal wetlands and marine sediments)
(第六届伯杰氏国际系统微生物学研讨会)

3. Oral presentation at the **"Xiamen Symposium on Marine Environmental Sciences"**. Xiamen City. 2025.01. (Title: *Candidatus* Nitrosomaritimum: an overlooked genus of ammonia-oxidizing archaea in intertidal aquifers and marine sediments).
(厦门海洋环境开放科学大会分会场口头报告)

4. Oral presentation at the **“2024 Annual Academic Meeting of the Chinese Association of Microbial Ecology”**. Qingdao City. 2024.10. (Title: *Candidatus* Nitrosomaritimum: an overlooked genus of ammonia-oxidizing archaea in intertidal aquifers and marine sediments) (中国生态学学会微生物生态委员会 2024 年学术年会分会场口头报告)
5. Oral Presentation in a sub-session of **“19th International Symposium on Microbial Ecology (ISME19)”**, Cape Town, South Africa, 2024.08. (Title: Ecogenomics and biogeochemical metabolism of a novel genus *Candidatus* Nitrosomaritimum widespread in anoxic saltmarsh intertidal aquifer)
6. **第十二届地质微生物学学术研讨会**研究生会场口头报告. 湖北武汉. 2024.04。(题目：滨海盐沼含水层发现的新型氨氧化古菌属 *Candidatus* Nitrosomaritimum)
7. **西湖大学可持续发展与前沿交叉论坛**海报展示(获优秀海报奖). 浙江杭州. 2024.03. (题目：Hydrodynamic-induced microbiota rhythmicity and assembly patterns in the intertidal aquifer and the key role of archaeal ammonia-oxidizers)
8. **第二十四次全国微生物学学术研讨会**海报展示. 贵州贵阳. 2023.08. (题目： Hydrodynamic and anthropogenic disturbances co-shape microbiota rhythmicity and community assembly within intertidal groundwater-surface water continuum)
9. **第十六届全国环境博士生学术会议**口头报告. 北京. 2022.10. (题目： Hydrodynamic and anthropogenic disturbances co-shape microbiota rhythmicity and community assembly within intertidal groundwater-surface water continuum)

ACADEMIC SERVICE

Peer Reviewer for: Journal of Environmental Science, Environmental Research.